

Fourier Tower Transforms and Quantum Circuits for Solvable Symmetries

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Abstract

We introduce a class of layered quantum circuits associated with solvable finite groups, called *Fourier tower circuits*. Rather than directly decomposing the group algebra into irreducible representations, the construction exploits a normal series decomposition of the group.

Each quotient layer is equipped with an abelian Fourier transform, while interactions between layers are implemented through conjugation-induced symmetry actions represented by controlled unitary gates.

The resulting architecture defines a hierarchical propagation of Fourier information across subgroup levels and provides a structured intermediate transform adapted to solvable-group symmetries.

We illustrate the construction explicitly for the groups S_3 and S_4 , analyze the associated controlled symmetry structure, and discuss connections with semidirect products, induced representations, and nonabelian quantum Fourier transforms.

1 Introduction

Quantum Fourier transforms over finite groups play a central role in quantum algorithms, notably in hidden subgroup problems and symmetry-based quantum computation.

For finite abelian groups, the Fourier transform admits efficient quantum implementations based on one-dimensional characters. The nonabelian case is considerably more difficult because irreducible representations become matrix-valued and the representation structure is substantially more complicated.

Standard nonabelian quantum Fourier transforms attempt to directly decompose the group algebra into irreducible representation sectors:

$$\mathbb{C}[G] \cong \bigoplus_{\rho \in \hat{G}} V_{\rho} \otimes V_{\rho}^*.$$

In this work, we propose a different perspective. Instead of directly diagonalizing the full group algebra, we exploit the subgroup structure of solvable groups through normal series decompositions. The key idea is to:

- decompose the group into successive quotient layers,
- apply local Fourier transforms on abelian quotients,
- couple neighboring layers through conjugation-induced symmetry actions.

This leads to a layered quantum-circuit architecture that we call a *Fourier Tower*.

The construction should not be viewed as a replacement for the full nonabelian quantum Fourier transform. Instead, it defines a structured intermediate transform adapted to solvable-group symmetries.

Remark 1. *The Fourier Tower construction may be viewed as a hierarchical generalization of the elementary identity*

$$HXH = Z,$$

where the Hadamard operator H is the Fourier transform on the group $\mathbb{Z}_2$.

In this elementary situation, the Fourier transform converts the translation operator X into the diagonal phase operator Z .

The Fourier Tower extends this principle from a single abelian symmetry layer to multiple quotient layers coupled through conjugation-induced symmetry actions.

2 Normal Series and Quotient Layers

Let G be a finite group.

Definition 1 (Normal Series). *A normal series of G is a chain of subgroups*

$$\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_n = G,$$

where each subgroup G_i is normal in G_{i+1} .

The associated quotient groups are

$$Q_i = G_{i+1}/G_i.$$

Each quotient defines one structural layer of the group.

If G is solvable, every quotient group Q_i is abelian.

This property allows one to associate an abelian Fourier transform with each layer independently.

2.1 Tower Coordinates and Transversal Sets

Definition 2 (Transversal Set). *Let*

$$H \triangleleft G.$$

A transversal set for the quotient

$$G/H$$

is a subset

$$T \subset G$$

containing exactly one representative from each coset of H in G .

Once such a transversal T is fixed, every element of G may be uniquely written as

$$g = th,$$

with

$$t \in T, \quad h \in H.$$

For the subgroup tower

$$\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_n = G,$$

we choose transversal sets

$$T_i \subset G_{i+1}$$

for the quotients

$$G_{i+1}/G_i.$$

Then every element

$$g \in G$$

admits a decomposition

$$g = t_{n-1}t_{n-2} \cdots t_0,$$

with

$$t_i \in T_i.$$

These coordinates define a hierarchical description of group elements compatible with the subgroup tower.

Remark 2. *The tower decomposition*

$$g = t_{n-1}t_{n-2} \cdots t_0$$

induces a natural factorization of basis states:

$$|g\rangle = |t_{n-1}\rangle \otimes \cdots \otimes |t_0\rangle.$$

Consequently,

$$\mathbb{C}[G] \simeq \bigotimes_i \mathbb{C}[Q_i].$$

This tensor decomposition provides the underlying Hilbert-space structure of the Fourier Tower circuit.

2.2 Example: Tower Coordinates for S_3

The symmetric group S_3 admits the solvable decomposition

$$\{e\} \triangleleft A_3 \triangleleft S_3,$$

with

$$A_3 \cong \mathbb{Z}_3, \quad S_3/A_3 \cong \mathbb{Z}_2.$$

Choose transversal representatives

$$T_0 = A_3 = \{e, (123), (132)\},$$

and

$$T_1 = \{e, (12)\}.$$

Then every element

$$g \in S_3$$

admits a unique decomposition

$$g = t_1 t_0,$$

with

$$t_0 \in A_3, \quad t_1 \in T_1.$$

For example,

$$(12)(123) = t_1 t_0,$$

with

$$t_1 = (12), \quad t_0 = (123).$$

Thus basis states factorize as

$$|g\rangle = |t_1\rangle \otimes |t_0\rangle.$$

This induces the tensor decomposition

$$\mathbb{C}[S_3] \simeq \mathbb{C}[\mathbb{Z}_2] \otimes \mathbb{C}[\mathbb{Z}_3].$$

3 Fourier Decomposition on Abelian Layers

Let Q_i be an abelian quotient group.

Its irreducible representations are one-dimensional characters

$$\chi : Q_i \rightarrow \mathbb{C}^\times.$$

The group algebra decomposes as

$$\mathbb{C}[Q_i] \cong \bigoplus_{\chi \in \widehat{Q_i}} \mathbb{C}_\chi.$$

Each character therefore defines a one-dimensional Fourier mode.

Definition 3 (Fourier Layer). *The decomposition*

$$\mathbb{C}[Q_i] \cong \bigoplus_{\chi \in \widehat{Q_i}} \mathbb{C}_\chi$$

is called the Fourier layer associated with the quotient group Q_i .

The local Fourier transform on the quotient layer is

$$F_{Q_i} : \mathbb{C}[Q_i] \rightarrow \bigoplus_{\chi \in \widehat{Q_i}} \mathbb{C}_\chi.$$

For cyclic groups

$$Q_i \cong \mathbb{Z}_n,$$

this reduces to the standard discrete Fourier transform.

4 Conjugation Actions Between Layers

The essential nonabelian structure of the Fourier Tower does not arise from the local Fourier transforms themselves, but from the interactions between layers.

Let

$$g \in G_{i+1}.$$

Conjugation induces an action on the quotient group:

$$g \cdot (xG_i) = gxg^{-1}G_i.$$

This further induces an action on characters:

$$(g \cdot \chi)(x) = \chi(g^{-1}xg).$$

Hence,

$$G_{i+1} \curvearrowright \widehat{Q_i}.$$

Thus higher layers act on lower Fourier modes through symmetry permutations.

Remark 3. *Without conjugation actions, the Fourier Tower would reduce to a simple tensor product of independent abelian Fourier transforms.*

The conjugation operators are therefore responsible for the nonabelian interaction structure of the circuit.

5 Tower Actions and Intertwining Structure

Let

$$\mathcal{H}_i = \mathbb{C}[Q_i]$$

be the Hilbert space associated with the quotient layer Q_i .

Definition 4 (Tower Action). *The conjugation-induced action*

$$\alpha_i : G_{i+1} \rightarrow \text{Aut}(\mathcal{H}_i)$$

is called the tower action on the layer $\mathcal{H}_i$.

The conjugation action defines the abstract symmetry structure between quotient layers, while the tower action provides its unitary realization on the associated Fourier-layer Hilbert spaces.

Remark 4. *The intertwining operator*

$$\mathcal{I}_i$$

ensures compatibility between neighboring symmetry layers.

If adjacent layers carry representations

$$\rho_i \quad \text{and} \quad \rho_{i+1},$$

then the intertwiner satisfies

$$\mathcal{I}_i \rho_i(g) = \rho_{i+1}(g) \mathcal{I}_i.$$

Thus symmetry information propagates coherently through the Fourier Tower.

6 Precise Structure of the Fourier Tower Transform

6.1 Tensor Decomposition of the Group Algebra

The tower coordinates induce a tensor decomposition:

$$\mathbb{C}[G] \simeq \bigotimes_{i=0}^{n-1} \mathbb{C}[Q_i].$$

The ordering of tensor factors follows the subgroup hierarchy from lower quotient layers to higher quotient layers.

6.2 Local Fourier Transform

For each quotient layer define

$$F_{Q_i} : \mathbb{C}[Q_i] \rightarrow \bigoplus_{\chi \in \widehat{Q_i}} \mathbb{C}_\chi.$$

The local Fourier transform is

$$F_{\text{loc}} = \bigotimes_i F_{Q_i}.$$

6.3 Controlled Conjugation Operators

Conjugation induces automorphisms

$$\sigma_y : Q_i \rightarrow Q_i.$$

Define permutation operators

$$P_y |x\rangle = |\sigma_y(x)\rangle.$$

The corresponding controlled symmetry operator is

$$C_i = \sum_{y \in Q_{i+1}} |y\rangle \langle y| \otimes P_y.$$

Thus,

$$C_i : |y\rangle |x\rangle \mapsto |y\rangle |\sigma_y(x)\rangle.$$

6.4 Global Fourier Tower Transform

The complete Fourier Tower transform is

$$F_{\text{tower}} = \left(\prod_i C_i \right) \left(\bigotimes_i F_{Q_i} \right).$$

The transform first performs independent Fourier decompositions on quotient layers and then couples them through controlled conjugation actions.

7 Relation with Semidirect Products

The Fourier Tower construction is naturally related to semidirect-product structures.

Suppose

$$G = N \rtimes H,$$

where H acts on N by conjugation:

$$h \cdot n = hnh^{-1}.$$

This is precisely the mechanism encoded by the controlled symmetry gates of the Fourier Tower.

7.1 Semidirect Structure of S_3

The group S_3 admits the decomposition

$$S_3 \cong \mathbb{Z}_3 \rtimes \mathbb{Z}_2.$$

The quotient layer $\mathbb{Z}_2$ acts on $\mathbb{Z}_3$ by inversion:

$$(12)(123)(12) = (132).$$

At the Fourier level this exchanges the nontrivial characters:

$$\chi_1 \leftrightarrow \chi_2.$$

7.2 Semidirect Structure of S_4

The group S_4 admits the decomposition

$$S_4 \cong (V_4 \rtimes \mathbb{Z}_3) \rtimes \mathbb{Z}_2.$$

The quotient

$$A_4/V_4 \cong \mathbb{Z}_3$$

acts cyclically on the nontrivial elements of V_4 :

$$(12)(34) \mapsto (13)(24) \mapsto (14)(23) \mapsto (12)(34).$$

At the Fourier level,

$$\chi_a \mapsto \chi_b \mapsto \chi_c \mapsto \chi_a.$$

8 Fourier Tower Circuit

The total Hilbert space is

$$\mathcal{H} = \bigotimes_i \mathcal{H}_i.$$

Definition 5 (Fourier Tower Gate). *A Fourier Tower gate is a unitary operator*

$$U_i = F_{Q_i} \circ \mathcal{I}_i \circ \alpha_i.$$

Definition 6 (Fourier Tower Circuit). *The global Fourier Tower circuit is*

$$U_G = U_n U_{n-1} \cdots U_1.$$

9 Explicit Circuit for S_4

The symmetric group S_4 admits the subgroup tower

$$\{e\} \triangleleft V_4 \triangleleft A_4 \triangleleft S_4.$$

The corresponding quotient layers are

$$V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2, \quad A_4/V_4 \cong \mathbb{Z}_3, \quad S_4/A_4 \cong \mathbb{Z}_2.$$

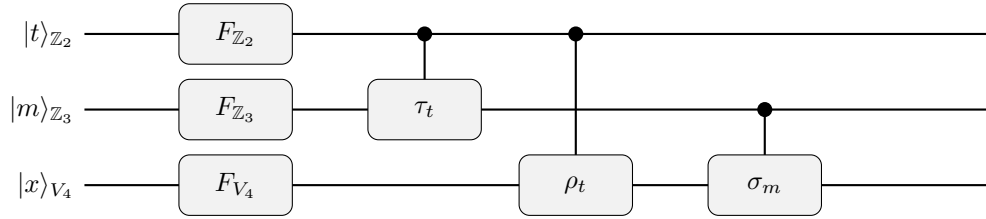
Thus the Fourier Tower circuit has three wires:

$$|x\rangle_{V_4}, \quad |m\rangle_{\mathbb{Z}_3}, \quad |t\rangle_{\mathbb{Z}_2}.$$

The local Fourier gates are

$$F_{V_4}, \quad F_{\mathbb{Z}_3}, \quad F_{\mathbb{Z}_2}.$$

The controlled gates encode the conjugation actions between the quotient layers. The Fourier Tower circuit for S_4 is:



Here:

- σ_m encodes the cyclic action of $\mathbb{Z}_3$ on the nontrivial Fourier modes of V_4 ,
- τ_t encodes the action of the top $\mathbb{Z}_2$ layer on the $\mathbb{Z}_3$ layer,
- ρ_t encodes the action of the top $\mathbb{Z}_2$ layer on the V_4 layer.

10 Main Theorem

Theorem 1 (Existence of Fourier Tower Circuits). *Let*

$$\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_n = G$$

be a normal series of a finite solvable group with abelian quotient groups

$$Q_i = G_{i+1}/G_i.$$

Then one can construct a layered unitary transform

$$F_{\text{tower}} = \left(\prod_i C_i \right) \left(\bigotimes_i F_{Q_i} \right).$$

Proof. Since G is solvable, it admits a normal series whose quotient groups are abelian.

Each quotient group Q_i therefore admits an abelian Fourier transform F_{Q_i} .

The tower-coordinate decomposition identifies the group algebra $\mathbb{C}[G]$ with a tensor product of quotient-layer Hilbert spaces.

Conjugation induces automorphisms between neighboring quotient layers. These automorphisms can be implemented as controlled permutation operators C_i .

Composing the local Fourier transforms with the controlled conjugation operators gives the Fourier Tower transform

$$F_{\text{tower}} = \left(\prod_i C_i \right) \left(\bigotimes_i F_{Q_i} \right).$$

□

11 Representation-Theoretic Interpretation of the Fourier Tower

11.1 Symmetry Group of an Operator

Let

$$U \in \mathcal{U}(\mathcal{H})$$

be a unitary operator.

Suppose a finite group

$$G_U$$

acts on $\mathcal{H}$ through a unitary representation

$$\rho : G_U \rightarrow \mathcal{U}(\mathcal{H}).$$

Assume

$$U\rho(g) = \rho(g)U.$$

Then G_U defines a symmetry group of the operator U .

If G_U is solvable, one may construct the associated Fourier Tower transform:

$$F_{\text{tower}}.$$

The Fourier representation of the operator becomes

$$\tilde{U} = F_{\text{tower}} U F_{\text{tower}}^{-1}.$$

Rather than directly diagonalizing the operator, the Fourier Tower organizes its symmetry structure hierarchically across quotient layers.

The Fourier Tower may be interpreted as a hierarchical version of this decomposition principle. Each quotient layer performs a partial Fourier decomposition, while the controlled conjugation gates implement the compatibility relations between neighboring layers.

11.2 Example: Fourier Representation of the CNOT and Toffoli Gates

The Fourier Tower philosophy may already be observed in elementary quantum gates such as the CNOT and Toffoli gates.

The essential principle is that Fourier transforms convert translation symmetries into phase symmetries.

11.2.1 The CNOT Gate

Consider the controlled-NOT gate

$$CX : \mathbb{C}^2 \otimes \mathbb{C}^2 \rightarrow \mathbb{C}^2 \otimes \mathbb{C}^2,$$

defined by

$$CX|a, b\rangle = |a, b + a\rangle, \quad a, b \in \mathbb{Z}_2.$$

The target transformation is

$$b \mapsto b + a.$$

Define the translation operator

$$X|b\rangle = |b + 1\rangle.$$

Then the CNOT gate may be written as

$$CX = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X.$$

Thus the symmetry group associated with the target action is

$$G_U = \mathbb{Z}_2.$$

The Fourier transform on $\mathbb{Z}_2$ is the Hadamard operator

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The fundamental Fourier identity is

$$HXH = Z.$$

Applying the Fourier transform on the target qubit gives

$$\tilde{U} = (I \otimes H)CX(I \otimes H).$$

Using

$$HXH = Z,$$

we obtain

$$\tilde{U} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes Z = CZ.$$

Explicitly,

$$CZ|a, b\rangle = (-1)^{ab}|a, b\rangle.$$

Therefore the Fourier transform converts the controlled translation symmetry into a controlled phase symmetry.

11.2.2 The Toffoli Gate

Consider now the Toffoli gate

$$\text{CCX} : (\mathbb{C}^2)^{\otimes 3} \rightarrow (\mathbb{C}^2)^{\otimes 3},$$

defined by

$$\text{CCX}|a, b, c\rangle = |a, b, c + ab\rangle, \quad a, b, c \in \mathbb{Z}_2.$$

The target transformation is again a translation action on

$$\mathbb{Z}_2.$$

Indeed,

$$\text{CCX} = \sum_{a,b \in \mathbb{Z}_2} |a, b\rangle \langle a, b| \otimes X^{ab}.$$

Applying the Fourier transform on the target qubit gives

$$\tilde{U} = (I \otimes I \otimes H) \text{CCX} (I \otimes I \otimes H).$$

Using again

$$H X H = Z,$$

we obtain

$$\tilde{U} = \sum_{a,b} |a, b\rangle \langle a, b| \otimes Z^{ab}.$$

This is the controlled-controlled- Z gate:

$$\tilde{U} = \text{CCZ}.$$

Explicitly,

$$\text{CCZ}|a, b, c\rangle = (-1)^{abc}|a, b, c\rangle.$$

Thus the Toffoli gate is transformed from a controlled-controlled translation operator into a controlled-controlled phase operator.

11.3 Example: Fourier Representation on the Nonabelian Group S_3

We now consider a genuinely nonabelian example illustrating how the Fourier Tower framework extends beyond elementary qubit gates.

The symmetric group

$$S_3 = \langle r, s \mid r^3 = e, s^2 = e, srs = r^{-1} \rangle$$

admits the solvable decomposition

$$\{e\} \triangleleft A_3 \triangleleft S_3,$$

with

$$A_3 \cong \mathbb{Z}_3, \quad S_3/A_3 \cong \mathbb{Z}_2.$$

Thus the Fourier Tower consists of:

- a Fourier layer associated with $\mathbb{Z}_3$,
- a second layer associated with $\mathbb{Z}_2$,
- and a conjugation action coupling both layers.

11.3.1 Fourier Layer on $\mathbb{Z}_3$

The irreducible characters of $\mathbb{Z}_3$ are

$$\chi_k(r^m) = \omega^{km}, \quad k, m \in \{0, 1, 2\},$$

where

$$\omega = e^{2\pi i/3}.$$

The Fourier transform on $\mathbb{Z}_3$ is

$$F_{\mathbb{Z}_3}|m\rangle = \frac{1}{\sqrt{3}} \sum_{k=0}^2 \omega^{km} |k\rangle.$$

The Fourier basis diagonalizes translations on the cyclic subgroup generated by

$$r = (123).$$

11.3.2 Conjugation Symmetry

The nontrivial element

$$s = (12)$$

acts on A_3 by conjugation:

$$sr s^{-1} = r^{-1}.$$

At the Fourier level, this induces an action on characters:

$$\chi_k \mapsto \chi_{-k}.$$

Explicitly,

$$\chi_1 \leftrightarrow \chi_2, \quad \chi_0 \mapsto \chi_0.$$

Therefore the $\mathbb{Z}_2$ quotient layer acts as a permutation symmetry on the Fourier modes of the $\mathbb{Z}_3$ layer.

11.3.3 Controlled Symmetry Operator

Define the controlled conjugation operator

$$C : |a\rangle|k\rangle \mapsto |a\rangle|\sigma_a(k)\rangle,$$

where

$$a \in \mathbb{Z}_2, \quad k \in \mathbb{Z}_3,$$

and

$$\sigma_0(k) = k, \quad \sigma_1(k) = -k \pmod{3}.$$

Thus the upper quotient layer dynamically permutes the Fourier modes of the lower layer.

11.3.4 Fourier Tower Representation

The total Fourier Tower transform for S_3 is

$$F_{\text{tower}} = C(F_{\mathbb{Z}_2} \otimes F_{\mathbb{Z}_3}).$$

The first step performs independent Fourier decompositions on the quotient layers. The second step introduces nonabelian interactions through conjugation-induced permutations.

Remark 5. *This example exhibits the central mechanism of the Fourier Tower construction:*

- *local quotient layers behave like ordinary abelian Fourier transforms,*
- *nonabelian structure appears through controlled conjugation actions,*
- *higher subgroup levels dynamically reorganize lower Fourier modes.*

11.3.5 Comparison with the Full Nonabelian Fourier Transform on S_3

The full nonabelian Fourier transform on S_3 decomposes:

$$\mathbb{C}[S_3] \cong V_{\text{triv}} \oplus V_{\text{sign}} \oplus V_{\text{std}} \otimes V_{\text{std}}^*,$$

where V_{triv} is the trivial representation, V_{sign} is the sign representation, and V_{std} is the two-dimensional standard representation.

The Fourier Tower does not directly construct these irreducible blocks.

Instead, it builds a hierarchical intermediate representation based on:

- quotient Fourier layers,
- semidirect-product symmetry actions,
- and controlled character permutations.

The standard representation of S_3 emerges from the interaction between the $\mathbb{Z}_3$ Fourier layer and the $\mathbb{Z}_2$ conjugation symmetry.

11.4 Relation with the Full Nonabelian Fourier Transform

The full nonabelian Fourier transform decomposes the group algebra as

$$\mathbb{C}[G] \cong \bigoplus_{\rho \in \widehat{G}} V_{\rho} \otimes V_{\rho}^*.$$

The transform maps basis states

$$|g\rangle$$

to representation coordinates

$$|\rho, i, j\rangle.$$

Explicitly,

$$F_G |g\rangle = \bigoplus_{\rho \in \widehat{G}} \sqrt{\frac{d_{\rho}}{|G|}} \sum_{i,j} \rho(g)_{ij} |\rho, i, j\rangle.$$

By contrast, the Fourier Tower does not directly separate all irreducible representations.

Instead, it organizes Fourier information hierarchically through quotient layers and conjugation-induced interactions.

Thus the Fourier Tower may be viewed as:

- a hierarchical Fourier decomposition,
- a solvable-group-adapted transform,
- and a structured intermediate representation between computational and fully nonabelian Fourier bases.

12 Complexity Considerations

Proposition 1. *Let G be solvable with quotient layers Q_i . Then the Fourier Tower circuit has complexity*

$$O\left(\sum_i \text{cost}(F_{Q_i}) + \sum_i \text{cost}(C_i)\right).$$

If the quotient groups are cyclic,

$$Q_i \cong \mathbb{Z}_{n_i},$$

then each local Fourier transform admits complexity

$$O(\log^2 n_i).$$

The remaining cost arises from controlled conjugation operators.

13 Conclusion

We introduced Fourier Tower quantum circuits associated with solvable finite groups.

The construction combines:

- local abelian Fourier transforms,
- quotient-layer decompositions,
- conjugation-induced symmetry actions,
- and controlled inter-layer operations.

This framework provides a structured symmetry-adapted approach to quantum circuit design and offers a new perspective on hierarchical Fourier analysis for solvable groups.

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